

Student Performance Prediction Using Machine Learning

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1. Introduction

Student performance prediction is an important application of Machine Learning in the education sector. Educational institutions can use predictive models to identify students who may require additional support and improve academic outcomes through data-driven decision making.

2. Project Objective

The primary objective of this project is to develop a Machine Learning model capable of predicting student performance based on academic and related factors. The project aims to demonstrate how predictive analytics can assist educational institutions in improving student success rates.

3. Problem Statement

Educational institutions often struggle to identify students who are at risk of poor academic performance. By analyzing historical student data, Machine Learning can generate predictions that help educators take proactive measures and provide targeted support.

4. TECHNOLOGIES USED

Python

Pandas

NumPy

Scikit-learn

FastAPI

Next.js

Streamlit

Machine Learning

Data Analysis

5. Methodology

The project follows a structured Machine Learning workflow. Data collection and preprocessing were performed to prepare the dataset. Relevant features were selected and transformed for model training. A Machine Learning algorithm was then trained, tested, and evaluated using standard performance metrics. The final model was integrated into a user-friendly application.

6. Model Development

The dataset was divided into training and testing sets. The model was trained on historical student records and optimized to generate accurate predictions. Various evaluation techniques were used to measure model performance and ensure reliability.

7. Results and Findings

The developed model successfully predicts student performance based on the provided inputs. The project demonstrates the practical application of Machine Learning in educational analytics and highlights how data can support informed decisions.

8. Business and Educational Impact

The solution can help institutions identify students who may require academic support, improve retention rates, optimize learning strategies, and enhance overall educational outcomes through predictive insights.

9. Project Links

GitHub Repository:	https://github.com/2403a52007/Student-Performance-Ai
Live Application:	https://student-performance-ai-jvba3ljx7nbwrc4thrpkgz.streamlit.app/

10. Conclusion

This project demonstrates the power of Machine Learning in solving real-world educational challenges. By combining data analysis, predictive modeling, and modern web technologies, the system provides a practical solution for student performance prediction while showcasing skills in AI, Machine Learning, and software development.